

Qingxi Meng

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Third-year Ph.D. student in Computer Science at Rice University with experience in perception, robotics, machine learning, and large-scale data compression. Skilled in Python, C++, PyTorch, and TensorFlow.

Education

Rice University

Ph.D., Computer Science

Advised by Profs. Lydia E. Kavradi and Vaibhav V. Unhelkar.

Houston, TX

2023 – Present

Stanford University

M.S., Electrical Engineering

Advised by Profs. Tsachy Weissman, Shreyas Vasanawala, and John Pauly.

Stanford, CA

2020 – 2022

University of Illinois at Urbana-Champaign

B.S., Electrical Engineering

Graduated with highest honors (GPA 4.0/4.0). Recipient of Robert M. Janowiak and TIA Foundation Scholarship.

Urbana, IL

2016 – 2020

Work Experience

Amazon AWS

Applied Scientist Intern (supervised by Dr. Dmitri Pavlichin)

Palo Alto, CA

Jun 2023 – Aug 2023

- Developed compression methods for pretrained transformer model parameters in AWS S3.
- Improved efficiency of large-scale ML storage and deployment pipelines.
- Reduced infrastructure costs by optimizing parameter storage and transmission.
- Skills:** Large-scale ML systems, transformer architectures, AWS S3, distributed computing, Python, C++.

Amazon AWS

Applied Scientist Intern (supervised by Dr. Dmitri Pavlichin)

Palo Alto, CA

Jun 2022 – Jan 2023

- Prototyped algorithms to reduce storage costs for transformer architectures.
- Designed experimental pipelines to evaluate storage-inference trade-offs.
- Collaborated with scientists to integrate prototypes into scalable ML infrastructure.
- Skills:** Model compression, Python, PyTorch, cloud infrastructure.

Philips Research North America

Research Intern (supervised by Dr. Patrick Cheung)

Cambridge, MA

Jun 2021 – Sep 2021

- Led project on compressing genomic sequencing quality scores using neural compression.
- Designed and trained deep learning models in PyTorch/TensorFlow for genomic data.
- Evaluated scalability of neural compression methods on large biomedical datasets.
- Skills:** Python, PyTorch, TensorFlow, C++, bioinformatics.

Research Experience

Rice University

Research Assistant

Houston, TX

Aug 2023 – Present

- Developing perception-aware motion planning for high-degree-of-freedom robots, combining motion planning with object and human detection tasks.
- Designed GPU-accelerated PRM/A* pipelines that incorporate perception costs from neural surrogate models.
- Conducted experiments on robotic tutors for nurse training, focusing on maintaining robust visual perception of humans and objects in dynamic environments.
- A paper accepted at *RSS 2025 [5]*; poster presented at *ICRA 2024 Workshop on Nursing Robotics [8]*.
- Skills:** 3D perception, scene graph, motion planning, GPU-accelerated algorithms, ROS/Isaac Sim.

Stanford University

Research Assistant

Stanford, CA

2021 – 2022

- Proposed neural network-based coil compression for MRI, enabling efficient representation of high-dimensional imaging data (*CVPR 2022 [6]*, *ISMRM 2022 [7]*).
- Developed NanoSpring for genomic data compression, integrating signal processing with learned models to reduce redundancy (*Scientific Reports 2023 [3]*).
- Skills:** Generative AI, image reconstruction, medical imaging, neural implicit model, PyTorch, TensorFlow.

Skills

Programming: Python, C++, C, SystemVerilog

Machine Learning: PyTorch, TensorFlow; experience with deep learning (CNNs, RNNs, transformers, diffusion models), generative models (GANs, VAEs), reinforcement learning, and optimization

Computer Vision / Perception: OpenCV, YOLO, segmentation, tracking, 3D reconstruction, SLAM

Robotics / Autonomous Systems: ROS/ROS2, PyBullet, Isaac Sim; control systems, sensor fusion (LiDAR, RGB-D, IMU)

Other: Distributed training, cloud platforms (AWS, GCP), Docker, Git, data compression, large-scale data pipelines

Publications

Journal

- [1] **Q. Meng**, E. Flores, C. Quintero-Peña, P. Qian, Z. Kingston, S. K. Hamlin, V. V. Unhelkar, L. E. Kavraki. "Look as You Leap: Planning Simultaneous Motion and Perception for High-DOF Robots." *Under review*, arXiv:2509.19610 [cs.RO], 2025. <https://arxiv.org/abs/2509.19610>
- [2] B. Lau, A. Almeda, M. Schauer, M. McNamara, X. Bai, **Q. Meng**, M. Partha, S. Grimes, H. Lee, G. Heestand, H. Ji. "Single molecule methylation profiles of cell-free DNA in cancer with nanopore sequencing." *Genome Medicine* 15, 33 (2023).
- [3] **Q. Meng**, S. Chandak, Y. Zhu, T. Weissman. "NanoSpring: reference-free lossless compression of nanopore sequencing reads using an approximate assembly approach." *Scientific Reports*, 13(1), 2082 (2023).
- [4] **Q. Meng**, I. Ochoa, M. Hernaez. "GPress: a framework for querying General Feature Format (GFF) files and feature expression files in a compressed form." *Bioinformatics*, 36(18), 4810–4812 (2020).

Conference

- [5] P. Qian, F. Bajraktari, C. Quintero-Pena, **Q. Meng**, S. Hamlin, L. E. Kavraki, V. V. Unhelkar. "ASTRID: A Robotic Tutor for Nurse Training to Reduce Healthcare-Associated Infections." *Proceedings of Robotics: Science and Systems (RSS)*, Los Angeles, CA, USA, June 2025.
- [6] E. Cole, **Q. Meng**, J. Pauly, S. Vasanawala. "Learned Compression of High Dimensional Image Datasets." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, New Orleans, LA, USA, June 2022.
- [7] **Q. Meng**, E. Cole, A. Raina, J. Pauly, S. Vasanawala. "Improved Neural Network-Based Coil Compression." *Proceedings of the International Society for Magnetic Resonance in Medicine (ISMRM) Annual Meeting*, London, UK, May 2022.

Poster and Talks

- [8] "Perception-aware Planning for Robotics: Challenges and Opportunities." Poster at *2024 IEEE 40th Anniversary of the International Conference on Robotics and Automation (ICRA@40)*, Rotterdam, Netherlands, September 24, 2024.
- [9] "Monitoring Constraints for Robotic Tutors in Nurse Education: A Motion Planning Perspective." Talk at *IEEE International Conference on Robotics and Automation (ICRA) 2024*, Workshop on Nursing Robotics, Yokohama, Japan, May 2024.
- [10] "NanoSpring: reference-free lossless compression of nanopore sequencing reads using an approximate assembly approach." *International Conference on Intelligent Systems for Molecular Biology / European Conference on Computational Biology (ISMB/ECCB)*, Virtual, July 2021.

Professional Service

Co-organizer, International Symposium on Medical Robotics (ISMR) 2025 Workshop on Robotics for Nursing.

Reviewer for:

- IEEE International Conference on Robotics and Automation (ICRA)
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
- IEEE Robotics and Automation Letters (RA-L)
- IEEE International Symposium on Information Theory (ISIT)
- International Conference on Machine Learning (ICML)